

Evaluating design-stage manufacturability decisions against the resolution of production

Hüseyin Tayyer Canseven^{1*}

^{1*}Department of Electrical Engineering, LUT School of Energy Systems,
LUT University, Yliopistonkatu 34, Lappeenranta, 53850, Finland.

Corresponding author(s). E-mail(s): huseyin.canseven@lut.fi;

Abstract

Design-stage manufacturability decisions increasingly rest on simulation and machine learning, yet whether a prediction is fit to decide is rarely evaluated against production. A framework is proposed whose unit, the production floor, is the difference that drift and scatter produce between two batches of one design within a run. A rule returning meets, fails or trial is characterised by a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong, for each relation of a new design to the produced evidence: a variant, a new process setting or a new family. A procedure turns them into a guard band for single verdicts. Thirteen rules were evaluated at a 95 % conformance level on open data of 18 deep-drawn design–process alternatives of 500 parts each, mostly process settings on one tool, with matched simulations; one floor of draw-in is 0.05 mm. With a produced sibling the nearest produced setting decided within 3.4 floors; for a new process setting no rule decided closer than 8.5 floors; in two transfers to a new family none was safe closer than 16. Within a family the choice of rule mattered more than the relation, and requirements at a performance index of 1.33 lay inside every decisive distance. With six to nine produced settings per fit, the simulation, as used, helped only across families, and learned models added abstention rather than accuracy. The framework gives evaluations of AI-based manufacturability support a production-referenced standard.

Keywords: design for manufacturing, machine learning, design decision making, verification and validation, operating characteristic, deep drawing

047 1 Introduction

048 Whether a design alternative can be made to its requirements is decided long before
 049 it is made. In sheet-metal forming, as in most processes, the decision rests on
 050 a finite-element simulation ~~of the process~~ (Banabic, 2010): the ~~design~~ team simu-
 051 lates the alternative, compares the predicted characteristics with the requirements
 052 ~~on the drawing and either releases the design~~ and releases the alternative, changes
 053 it or sends it to a physical trial. Simulation has moved such decisions forward in
 054 the design process (~~Thomke, 1998; Thomke and Fujimoto, 2000~~) (Thomke, 1998), and
 055 machine learning ~~has made~~ makes the loop faster and ~~broader~~ surrogates broader:
 056 surrogates replace simulations, ~~data-driven models learn from earlier products, and~~
 057 ~~manufacturability checks are increasingly automated~~ (Yan et al, 2026; Liu et al, 2026)
 058 ~~—but neither models learn manufacturability from earlier designs, and checks~~
 059 ~~are automated~~ (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018). Neither has
 060 changed the question ~~a designer~~ the team has to answer: *can this prediction be* ~~trusted~~
 061 ~~relied on to decide?~~ A simulation that is right on average can ~~still~~ be wrong for the
 062 alternative at hand, and a model ~~that learned from~~ learned on earlier products may be
 063 asked about a design unlike any of them. Whitney argued in this journal that mechan-
 064 ical design cannot be verified by simulation the way integrated circuits are (Whitney,
 065 1996), ~~and variation in production remains the reason~~ (Thornton et al, 2000): ~~the~~
 066 ~~credibility of~~ production variation, with which development organisations still
 067 contend (Thornton et al, 2000), ~~adds a further reason. Whether~~ a simulation-based
 068 decision is ~~ultimately credible is in the end~~ a property of production, and production
 069 evidence is ~~exactly~~ what is scarce at the design stage.

070 This paper treats the design-stage manufacturability decision itself as the object
 071 of study and asks three questions. How can the fitness of a design-stage prediction for
 072 a manufacturability decision be expressed ~~independently of the predictor and of the~~
 073 ~~on a common production-referenced scale, for any predictor and~~ characteristic (RQ1)?
 074 How does that fitness depend on the production evidence ~~available—on available, on~~
 075 its amount and on how the new design relates to what has been produced (RQ2)? And
 076 what do simulation and machine learning contribute to such a decision once production
 077 evidence exists (RQ3)? The questions ~~sit between three strands of design research:~~
 078 ~~matter for~~ decision-based design ~~; which treats design as a sequence of decisions under~~
 079 ~~uncertainty~~ (Hazelrigg, 1998; Lewis et al, 2006); ~~(Hazelrigg, 1998; Lewis et al, 2006),~~
 080 ~~for~~ the margins that designers keep between predictions and requirements (Eckert
 081 et al, 2019; Brahma and Wynn, 2020) ~~; and and for~~ the planning of verification and
 082 testing ~~in product development~~ (Thomke and Bell, 2001; Salado and Kannan, 2018;
 083 Tahera et al, 2019). ~~Each~~ each needs to know how far a prediction can be ~~trusted~~
 084 ~~relied on~~ before a test is cheaper than a decision, ~~and none has a production-referenced~~
 085 ~~measure of it.~~

086 ~~A framework that provides such a measure is proposed. Its unit is~~ The framework
 087 ~~proposed here has~~ the *production floor as its unit*; the difference ~~that drift and scatter~~
 088 ~~produce~~ between two batches of consecutive parts of one ~~and the same design that~~
 089 ~~drift and scatter produce at a chosen confidence~~ design within a run, a floor in the
 090 sense of a noise floor rather than of the shop floor. A rule that turns the design-
 091 stage evidence into ~~one of three verdicts—the a verdict~~ (the alternative *meets* the
 092

requirement, *fails* it, or goes to a physical *trial*—~~is~~) is characterised by two operating
characteristics ~~expressed in floors: its *decisive distance*, beyond which at least 95 % of~~
~~its verdicts are correct, in floors, its *decisive distance* and its *safe distance*, beyond~~
~~which at most 5 % are wrong. Both are functions of how far the requirement lies~~
~~from the truth, in the way the operating characteristic of an acceptance plan is a~~
~~function of the true quality, and between them the rule abstains rather than errs.~~
~~They are evaluated separately for each *evidence relation* between a new design and~~
~~the produced alternatives—a variant whose process setting has been produced with~~
~~another, unresolvable variation; a new process setting within or outside the produced~~
~~range; and a new design family—together with a calibration budget, an applicability~~
~~guard and a six-step procedure~~alternatives, and a procedure turns them into a guard
band on the margin of a single verdict. The decisions in scope are those of a team that
has produced variants of a design family: the release of a further variant or process
setting in process planning and tryout, and the transfer of production knowledge to a
new family; early product design without production evidence of the family is covered
only by the new-family relation.

The framework is evaluated on open data that make ~~such an evaluation~~
this possible: 9,000 deep-drawn and cut cups of dual-phase steel, pro-
duced as 18 design-process alternatives of 500 consecutive parts each ~~;~~
~~scanned after drawing and after cutting (Baum and Heinzelmann, 2026), together~~
~~and scanned (Baum and Heinzelmann, 2026),~~ with finite-element simulations of
the same process ~~and tools (Baum and Heinzelmann, 2025).~~ Each produced
(Baum and Heinzelmann, 2025). Within each of two cup geometries the alternatives
differ in blank-holder force and lubrication on one tool, so two of the three relations
concern process settings, and the two geometries give two transfers between families.
Each alternative is judged as a new ~~design—by leaving it out alone, or with every~~
~~alternative that shares its process setting, or with its whole family—and design, and~~
the verdicts of ~~twelve rules~~ thirteen rules at a 95 % conformance level are compared
with what its 500 parts ~~actually~~ did. In the terms of the design research methodol-
ogy (Blessing and Chakrabarti, 2009), ~~the paper this~~ is a prescriptive study ~~whose~~
~~framework and procedure are given an initial evaluation on production data; that~~
~~evaluation establishes with an initial support evaluation: it shows~~ whether the frame-
work discriminates between rules and evidence situations, not ~~how design teams use~~
~~it whether design teams decide better with it, and its findings on one dataset are~~
hypotheses for confirmation.

The contributions are ~~fourfold. The first is the framework itself (Section 3): the~~
~~production floor with its effect-to-scatter ratio and minimum resolvable change, and~~
~~the decisive and safe~~ the framework, with the floor positioned as a precision limit of
~~batch centres and the~~ distances as operating characteristics ~~;~~ ~~with their assumptions,~~
~~properties and with~~ an exact estimator ~~;~~ ~~The second is (Section 3): an evaluation~~
~~design for design-stage decision rules~~ that separates the evidence relations by grouped
cross-validation (Roberts et al, 2017) and isolates what the simulation ~~contributes~~
adds by evaluating every calibrated rule with and without it (Section 4.4). ~~The third~~
~~is the empirical answer to the three questions on deep drawing, including what the~~
~~production floor contains and how the conclusions depend on the analysis choices~~

4); the empirical answers for deep drawing (Section 5). ~~The fourth is a procedure and guidance for design practice that separate what is general from what holds for this dataset~~; and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support elsewhere (Section 6).

Section 2 positions the work ; Section 3 defines the framework; Section 4 describes the data, the characteristics and the evaluation design; Section 5 reports the results; Section 6 discusses what they establish and the threats to validity; and Section 7 concludes. Appendix A gives computation details and additional tables summarises the notation, and the Electronic Supplementary Material (ESM) holds further results.

2 Background and related work

2.1 Manufacturability evaluation and machine learning ~~in~~ design for manufacturing

Design for manufacturing (DFM) ~~has long codified~~ codifies what a process can make into rules, guidelines and cost models applied at the design stage (~~Boothroyd et al, 2010; Bralla, 1999~~) (Boothroyd et al, 2010), and quantitative manufacturability models have been used in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Process simulation ~~now~~ predicts the outcome of a specific alternative ~~rather than of a class of features~~, and machine learning learns manufacturability from ~~historical designs and their production records, from CAD models~~ (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019) ~~from process data and from simulated design spaces, or from both~~ (Yan et al, 2026; Ballegeer and Benoit, 2026) (Yan et al, 2026; Ballegeer and Benoit, 2026), and generative models propose designs directly (Regenwetter et al, 2022). Work in this journal has used data and models to choose ~~between manufacturing processes from shape~~ (Ben Slama et al, 2024), ~~to quantify design robustness~~ (Li et al, 2026), ~~to reason about failure causes in design FMEA~~ (Liu et al, 2026) and to show that measures left out of an alternative-selection problem change which alternative is chosen (Eaton et al, 2026). ~~A recurring difficulty is transfer: a model learned on one process, machine or product family is applied to another~~ (Safdar et al, 2024), and its reproducibility and reusability are open questions (Xie et al, 2025, 2026). ~~What these studies rarely have is a production record~~. Transfer between processes, machines and product families is a recurring difficulty (Safdar et al, 2024). Learned manufacturability models are typically evaluated by held-out accuracy or classification scores on labelled designs (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), not by the decisions they support against the production of the very alternatives ~~the model is asked about, so their evaluation stops at prediction accuracy and does not reach the design decision~~ they are asked about; open datasets that would allow more are ~~increasingly recognised as research~~ recognised as contributions in their own right (Ahmed et al, 2025).

2.2 Design decisions ~~under uncertainty~~, design types and their ~~evaluation~~ production evidence

Decision-based design casts design as ~~choosing between alternatives whose outcomes are uncertain, with a preference over them (Hazelrigg, 1998; Lewis et al, 2006)~~. ~~In practice a choice between alternatives with uncertain outcomes (Hazelrigg, 1998; Lewis et al, 2006).~~ How much a team knows about a new alternative depends on the kind of design: variant design changes parameters of a proven solution, adaptive design changes parts of it and original design creates a new one (Pahl et al, 2007), and product-family and platform design organise these changes across products (Simpson et al, 2006; Jiao et al, 2007). The evidence relations of this paper are their counterparts for production evidence: a new variant of a produced setting, a new process setting of a produced design, and a new family. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999), key characteristics (Thornton, 1999) and tolerance analysis (Chase and Parkinson, 1991); a rule that returns the capability of the nearest produced setting is a capability-database look-up. Where the uncertainty of a prediction is not resolved it is absorbed by margins ~~:- buffers between what is predicted and what is required, which are seldom made explicit and whose value is rarely known (Eckert et al, 2019; Brahma and Wynn, 2020)~~. ~~Testing is (Eckert et al, 2019; Brahma and Wynn, 2020), and the part of a margin that covers uncertainty is distinct from the part that absorbs change (Eckert and Isaksson, 2017).~~ Testing and prototyping are the alternative to a margin, and its planning ~~when~~ when to test, how many alternatives to test ~~in parallel or in sequence, and which verification activities to perform has and in which order has~~ been modelled as a trade between the cost of tests and the cost of errors (Thomke, 1998; Thomke and Bell, 2001; Loch et al, 2001; Salado and Kannan, 2018) ~~, with physical and virtual testing (Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018)~~, with testing and prototyping strategies studied as design activities ~~in their own right (Tahera et al, 2019).~~ Variation risk management identifies the key characteristics whose variation matters and the processes that control them (Thornton, 1999; Thornton et al, 2000). The framework of this paper supplies a quantity these approaches assume but do not measure ~~:~~ how close to a requirement (Tahera et al, 2019; Camburn et al, 2017). When upstream information is fit to act on is the question of overlapped development and preliminary information (Krishnan et al, 1997; Terwiesch et al, 2002), and set-based concurrent engineering codifies production knowledge as feasibility limits for new designs (Sobek et al, 1999). All of these assume a measure of how far a design-stage prediction can ~~decide, be relied on; the framework supplies one~~ in a unit set by production, ~~and therefore how large a margin has to be, or when a test is unavoidable.~~ The evaluation of design methods themselves has its own methodology: the design research methodology separates the development of support from its evaluation (Blessing and Chakrabarti, 2009), the validation square distinguishes the structural and performance validity of a method (Seepersad et al, 2006), and evaluation designs from medicine have been proposed for design methods (Frey and Dym, 2006).

2.3 Simulation credibility, calibration and validation metrics

Verification and validation methodology asks whether a model is accurate enough for its intended use, and answers it by comparing predictions with experiments under a stated validation metric (Oberkampf and Roy, 2010; Roy and Oberkampf, 2011); standards for computational solid mechanics and for the credibility of decisions based on models formalise the same question (ASME, 2020; NASA, 2024). Validation metrics compare predicted and observed distributions in physical units (Ferson et al, 2008; Liu et al, 2011), and predictive maturity tracks how the predictive capability of a calibrated model improves with validation experiments (Hemez et al, 2010) (Oberkampf and Roy, 2010), formalised in risk-informed credibility assessment, which scales the required evidence with the consequence of a wrong decision (ASME, 2018). The distinction between the validation domain and the application domain of a model (Oberkampf and Roy, 2010) is the question that the evidence relation asks of a design. Bayesian calibration adds a model of the discrepancy between simulator and reality, typically a Gaussian process over the inputs, together with calibration parameters of the simulator (Kennedy and O'Hagan, 2001; Higdon et al, 2008; Bayarri et al, 2007); discrepancy model and calibration parameters to a simulator (Kennedy and O'Hagan, 2001); multi-fidelity models scale a cheap model to an expensive one (Kennedy and O'Hagan, 2000), and omitting the discrepancy biases the calibrated parameters (Brynjarsdóttir and O'Hagan, 2014), and the identifiability of discrepancy and parameters limits what few experiments can teach (Arendt et al, 2012) parameters (Brynjarsdóttir and O'Hagan, 2014). Where a predictive model may be trusted relied on at all is the question of its valid input domain (Malak and Paredis, 2010). Conformal prediction gives distribution-free intervals with a finite-sample coverage guarantee under exchangeability (Vovk et al, 2005; Lei et al, 2018; Barber et al, 2021; Angelopoulos and Bates, 2023) (Lei et al, 2018; Barber et al, 2021), including for quantiles (Romano et al, 2019) and under a shift of the inputs (Tibshirani et al, 2019), under covariate shift (Tibshirani et al, 2019) and for the control of decision risks (Angelopoulos et al, 2024), and it has begun to appear in engineering design and quality assurance (Zong et al, 2026; Heddoub et al, 2026). These methods produce intervals and accuracy statements; how close to a requirement such an interval can be trusted to decide, and how much production evidence that takes, remains open.

(Zong et al, 2026). In sheet-metal forming, the accuracy of finite-element predictions springback above all has been studied for decades (Wagoner et al, 2013; Chen et al, 2022) predictions has long been studied (Wagoner et al, 2013), with friction (Hol et al, 2012) and the unloading behaviour of the material (Yoshida and Uemori, 2002) among the main sources of error. Simulations are used to compensate springback in the tool design (Gan and Wagoner, 2004), robust process design under material and process scatter is well established (Strano, 2006; Hou et al, 2010; Wiebenga et al, 2014), often with metamodels (Jakumeit et al, 2005; Harsch et al, 2016), and data-driven models predict forming outcomes from process data in series production (Havinga et al, 2020; Purr et al, 2015). Simulation datasets of deep drawing

have been released for surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), ~~with surrogates trained on them~~ (Heinzelmann et al, 2025b; Baum et al, 2025) and the gap between synthetic and real data has been identified as the obstacle to ~~their~~ industrial use (Baum et al, 2026b). ~~The dataset’s authors decomposed, and~~ the deviation between simulation and parts ~~and found that the part geometry explains 77–92 % of its variance and that gradient boosting with process measurements explains 81–92 % (Baum et al, 2026a); the present paper asks the complementary design question of~~ has been decomposed statistically (Baum et al, 2026a). These methods produce intervals and accuracy statements; how close to a requirement ~~a decision taken before production, without process measurements of the new alternative, can be trusted~~ such an interval can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

~~Three fields evaluate decisions that may abstain.~~ In conformity assessment ~~under measurement uncertainty~~, a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty and the ~~specific and global~~ risks of false acceptance and false rejection as the measures of a decision rule (ISO, 2017b; Joint Committee for Guides in Metrology, 2012). In ~~analytical chemistry, the limit of detection is the smallest quantity a method detects with stated error rates (Currie, 1968, 1995), and in acceptance sampling and selection~~ selection procedures and acceptance sampling the operating characteristic gives the probability of a decision as a function of the true quality (Bechhofer, 1954). ~~In machine learning, a~~ with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970), summarised by risk-coverage curves (Geifman and El-Yaniv, 2017), ~~and interval forecasts are scored by their coverage and width (Gneiting and Raftery, 2007); learning to defer passes difficult cases to an expert (Madras et al, 2018); and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), which matters for design teams working with AI (Zhang et al, 2021).~~ The decisive and safe distances are operating characteristics of this kind for design-stage manufacturability decisions; ~~what the framework adds is a unit set by production, which makes them comparable across characteristics and predictors, and their evaluation by evidence relation, and the guard band of the procedure is the design-stage counterpart of a conformity-assessment guard band.~~

2.5 Production variation and measurement **capability**

Statistical quality control separates ~~the variation of a process into~~ short-term and long-term ~~components and compares it~~ variation and compares them with tolerances through capability ~~indices (Montgomery, 2019; Kane, 1986)~~ and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006)

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible on the measured item	yes
Conformity assessment, OC curves (ISO, 2017b; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks		yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

and time-dependent process models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5% (ISO, 2025), and measurement systems analysis separates repeatability from stability and decides whether a gauge can resolve a tolerance (Automotive Industry Action Group, 2010; ?) (Automotive Industry Action Group, 2010). The production floor is a ~~quantity of the same family, measured between batches of one design: it contains short-term scatter only through the batch average, and drift within a series in full.~~ repeatability-type limit of batch centres under within-run conditions; what is new is its use as the unit of a design decision rule. Table 1 places the framework among these approaches. ~~It~~ it does not replace calibration or conformal prediction ; ~~it uses them but scores them,~~ as decision rules ~~and scores them,~~ against production in a common unit, ~~which is what a design team needs to choose between them.~~

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts. A quality characteristic c varies over the parts of a series. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p -quantile of its parts satisfies $q_p(a, c) \leq R$. ~~The demonstration uses $p = 0.95$ and checks $p = 0.90$ and 0.99 ; requirements on a lower limit~~ Lower or two-sided limits follow by a change of sign or by the absolute deviation from the target; a tolerance of $\pm t$ on a wall angle about its design angle, for example, is the limit $R = t$ on the absolute deviation. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment ~~, in which uncertainty creates a zone where neither conformance nor non-conformance can be proven (ISO, 2017b); here the uncertainty is that of the (ISO, 2017b), here for the uncertainty of a~~ design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*. ~~The decisions in scope are those of a design team that has produced variants of a design family and has to decide on; it is used by the process planner or tryout engineer for a further variant; a new process setting or a new family before producing it; the release of variants and process settings in process planning and tryout; and the transfer of production knowledge to or process setting of an existing tool, at tool tryout or before production part approval, and by the product designer for a new family in platform and variant design.~~ The demonstration decides on $p = 0.95$, which for a normal characteristic corresponds to a performance index of about 0.55. Release criteria of series production, a performance index of 1.33 or more (Automotive Industry Action Group, 2006), concern nonconforming fractions in parts per million that a series of 500 parts cannot estimate without a parametric tail; the framework reaches them through requirements anchored in capability (Section 5.8).

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts. ~~Each series is,~~ each cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10% trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i < j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design, ~~pooled over the produced alternatives (here $B = 50$,~~. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2/B}$ at $\alpha = 0.95$ ~~and series of ten batches; floors per geometry are used;~~ the repeatability limit of precision experiments (ISO, 2025, 1994), applied to batch centres under within-run conditions. The reference protocol of this paper is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with ~~per-geometry floors where geometries differ~~). The floor rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); the runs from which it is estimated are like those of future production (A2); and the ~~measurement adds little to it~~ measuring system is repeatable and its resolution small against the floor (A3, checked in Section 5.1). ~~It has three properties that~~

Four properties matter for its use. It has the unit of the characteristic, so ~~a margin expressed in floors does not depend on the unit or the origin of the characteristic.~~ Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2/B}$ at $\alpha = 0.95$, so the short-term capability enters through $\sigma_w/\sqrt{B}$ while drift within a run enters in full; the floor obtained after the parts of each series are randomly permuted contains scatter only, and its ratio to the floor measures the drift. ~~And the floor distances in floors are on a common scale for all characteristics and predictors, although equal distances do not mean equal nonconforming fractions, because the ratio of the floor to the scatter of the parts differs between characteristics (Table 3).~~ It is a property of a production protocol: under drift it grows with the ~~length of the series under drift,~~ time between the batches compared, and it shrinks with B ; σ_b , σ_w and the floor at every lag are reported so that it can be recomputed for another protocol (ESM Tables S3 and S4). It contains whatever drifts between batches, the process and the measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it excludes the variation between runs, which ~~bounds it from below as a measure makes it~~ a lower bound of the variation of series production. ~~The~~ For a new family, whose floor is unknown before it is produced, the floor of the produced family is used. Because the floor resolves batch centres while the requirement concerns a quantile; ~~it is used as a unit, and,~~ the reproducibility of the quantile ~~itself~~ is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; ~~this is a practical threshold, not a test.~~ For a continuous design or process parameter x with a local sensitivity $s_c = \partial\bar{m}/\partial x$, the minimum resolvable change is $F_c/|s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local, a statement about centres; ~~,~~ and a change below it can still move the tail of the distribution.

3.3 Decision rules

~~Twelve–Thirteen~~ rules span what a design team can do with its evidence (Table 2);
each is defined for any set of calibration alternatives. ~~Two use the simulation alone;~~
~~the,~~ and the first seven carry the results in the main text. The nominal simulation
(M0) ~~and the worst case of a process-window simulation uses the simulation alone. The~~
~~bias-corrected simulation (M0w), as a robustness simulation is read in practice. Five~~
~~combine the simulation with the production record. M1 adds to the nominal simulation~~
) adds the median offset between the calibration alternatives’ q_p and their nominal sim-
ulation. ~~Mc calibrates a parameter of the simulator instead: the friction coefficient of~~
each lubrication pattern is chosen so that the simulated draw-in matches the produced
alternatives, in the sense of a calibration parameter of Kennedy and O’Hagan (2001)–
without a discrepancy model, and the median remaining offset is added.; ~~the scaled~~
simulation (M1s) fits $q_p = a + b s_0$ to them by least squares, a linear multi-fidelity
correction (Kennedy and O’Hagan, 2000) that rescales a simulated trend of the wrong
size. The envelope rule (M2 ~~takes the upper end of the simulation envelope over~~
~~plausible incoming conditions,~~) adds the median offset and ~~a margin equal to~~ the
largest absolute residual of the calibration alternatives ~~to the upper end of the~~
~~simulation envelope over plausible incoming conditions.~~ M5 models the offset between
 q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) ~~over~~
~~the alternatives’ descriptors, so that the correction may vary with the design and~~
~~process settings, and uses its~~ with a 90 % predictive interval. M3 is a part-level model:
gradient boosting (Friedman, 2001) ~~learns the discrepancy between the measured~~
~~characteristic and the matched simulation from the descriptors and the measured~~
~~incoming conditions of the calibration parts, and the predicted parts of the new~~
~~alternative give its q_p , with an offset and a margin from a leave-one-out over the~~
~~calibration alternatives (a jackknife) . Five rules use the production record alone~~
~~and isolate what the simulation adds: the median q_p of the calibration alternatives~~
(M1n) ; the same with the residual margin (M2n), the Gaussian process of , ~~whose~~
~~noise variance is bounded below by the variance between the produced series of one~~
~~geometry and force, so that a new series of a produced setting is not predicted~~
~~more precisely than produced series reproduce each other; M5n is the same process~~
~~of q_p itself (M5n), the without the simulation. The nearest produced setting (NN)~~
is a capability-database look-up that uses neither simulation nor learning. The six
further rules include a friction-calibrated simulation (Kennedy and O’Hagan, 2001)
(Mc) and part-level ~~boosting model of the characteristic (M4), and the nearest~~
~~produced setting (NN), which takes the mean q_p of the calibration alternatives of~~
~~the design’s family at the nearest process setting~~ gradient boosting (Friedman, 2001)
with a jackknife+ interval (Barber et al, 2021), with and without the simulation (M3,
M3n). The nominal simulation and the envelope do not depend on the lubrication
pattern, so a lubrication variant of a produced setting gives M0, M1, M1s, M2 and
M5 no simulated information that distinguishes it from its siblings.

The margins are labelled by what they guarantee. With n calibration alternatives,
the largest-residual margin ~~of M2 and M2n~~ covers a new exchangeable residual with
probability $n/(n + 1)$ ~~when the offset is fixed and for a fixed offset and at least~~ $(n -$
 $1)/(n + 1)$ ~~when it is estimated from the same for an estimated one, because the~~
new residual is then covered whenever it falls within the range of the calibration

Table 2 The decision rules; the first seven carry the results in the main text

Rule	Prediction of q_p	Interval	Production evidence
M0 nominal simulation	simulation at nominal sheet thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares on the calibration alternatives	point	q_p
M2 envelope + residual margin	upper end of the envelope plus the median offset	largest residual	q_p
M5 GP correction of the envelope	envelope plus a Gaussian-process offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	Gaussian process of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced process setting of the family	point	q_p
M0w process-window worst case	lowest to highest simulated value over the incoming conditions	range	none
Mc friction-calibrated simulation	simulation with friction calibrated on the draw-in, plus the median offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus a boosting model of the discrepancy, part level	jackknife+	parts
M3n boosting, no simulation	boosting model of the characteristic, part level	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). GP: noise variance bounded below by the variance between the series of one geometry and force. M3n is named M4 in the released code; M6, a conformalised GP, appears only in the robustness analysis.

residuals; it is the split-conformal quantile (Lei et al, 2018) at the highest ~~level that n residuals can attain~~ attainable level, not a 90 % interval. ~~The Gaussian-process interval is model-based, and the jackknife margin of M3~~ In general, a distribution-free interval cannot certify a miss rate below $1/(n+1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the relevant relation and one of 10 % needs nine. Under exchangeability the jackknife+ interval covers with probability at least 0.8 at the level of 0.9 used here (Barber et al, 2021), and the Gaussian-process interval is model-based. The level is a decision variable that trades decisiveness for safety, as the cost of errors would set the threshold of a reject option (Chow, 1970); 0.9 is used here, and 0.8 and M4 carries no guarantee (Barber et al, 2021). Their empirical coverage is reported in Section 5.50.95 are reported in the robustness analysis.

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d F_c, \quad (4)$$

so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and ~~wrong~~ (a false reject) when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and ~~wrong~~ (a false accept) when $-u \geq \delta$. ~~Over a population of cases, let~~ Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$, ~~both signs weighted equally and requirements below zero excluded for characteristics that cannot be negative; every case gives one verdict on an adequate alternative and, where $q_p - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal weight. As δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors (ESM Table S11).~~ The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$.

The ~~two distances have five properties. They~~ distances are exact: κ and ω change only where δ crosses one of the case's u , v , $-u$ or $-v$, so the distances are order statistics of these exclusion distances ~~and need no grid~~; for a rule that always decides, $\delta_d = \delta_s$ is close to the 90 % quantile of its absolute ~~prediction~~ error in floors ~~when no requirement is excluded~~. Always $\delta_s \leq \delta_d$, and between the two the rule sends designs to trial rather than misleading. ~~They do not depend on the unit of the characteristic. They split into one-sided distances. The distances are reported for both sides together and one-sided, for failing alternatives, where the errors are false accepts, and for adequate ones, where they are false rejects (false accepts) and adequate ones (false rejects), which matters when the two errors cost differently. And they are operating characteristics (ESM Table S14). They are points on operating characteristics, functions of the true margin over a population of design situations, like the operating characteristic of an acceptance plan (Bechhofer, 1954) or the risk coverage curve of a selective classifier (Geifman and El-Yaniv, 2017), not statements about a single verdict; the decisive distance is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95, and the pair plays the role of the acceptable and rejectable quality levels of an acceptance plan.~~ A design team uses them before a requirement is fixed, to choose a rule for a family and evidence relation, to judge whether a requirement ~~placed~~ near the expected capability can be decided at the design stage at all, and to plan trials and production evidence. ~~For a single verdict the observable quantity is the margin between the predicted interval and the requirement; the share of correct verdicts as a function of that margin is reported with the results.~~

~~In decision-theoretic terms, let a false accept cost c_{FA} , a false reject c_{FR} and a trial c_T , with correct verdicts free and trials A trial is itself not error-free. ~~Over the population at a true margin beyond δ_d , the expected cost of a verdict is at~~~~

most $0.05 \max(c_{FA}, c_{FR}, c_T)$, and beyond δ_s at most $0.05 \max(c_{FA}, c_{FR}) + u(\delta) c_T$ with $u(\delta)$ the share sent to trial; the bound holds on average over adequate and failing alternatives weighted equally, and one-sided distances replace it when the base rates differ. Which rule to prefer then depends on the cost ratios; a worked comparison with realistic requirement margins is given in Section ??; a short run reproduces the 95th percentile only to about one floor (Section 5.1), so no decision can be relied on closer to the requirement than the trial that would replace it.

3.5 Evidence relation, calibration budget and applicability guard

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced with another variation that production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); and a *new family* shares no produced alternative. The ~~two~~ distances are evaluated for each relation. Beyond a factorial, the relation can be graded by a distance in descriptor space or by the effect-to-scatter ratio between the predicted characteristic of the new design and its nearest produced alternative (Section 6.5). The calibration budget is ~~their dependence~~ the dependence of the distances on the number k of produced alternatives, over all subsets of size k ~~and the calibration design asks which alternatives to produce first; subsets are classified by the relation of~~ classified by their relation to the new design ~~to them~~. An applicability guard returns *trial* when the new design lies outside what the calibration has seen: its geometry is not among the calibration alternatives (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the range of the calibration alternatives' simulated values (novelty guard); ~~all three use design-stage information only~~, in the spirit of a valid input domain (Malak and Paredis, 2010).

3.6 Procedure

Figure 1 turns the framework into six steps. ~~Steps: steps 1 to 5 are carried out once per design family and, repeated when production evidence is added; step, and step 6 is carried out for every new design.~~ Step -1 frames the decision ~~: the family, its alternatives, the characteristics and the~~ (family, alternatives, characteristics, form of the requirement - Step and conformance level); step 2 measures the floor ~~on the produced alternatives, and step with a stated protocol;~~ step 3 states which differences between alternatives, and which local changes of a parameter, production can resolve; ~~a design question below the floor is not one the design stage needs to answer finer.~~ Step and step 4 calibrates the candidate rules, with and without the simulation, and step. Step 5 qualifies them by their ~~decisive and safe~~ distances for each evidence relation ~~, their calibration budget and the guard; when and fixes for each rule and relation a guard band g ; the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct, for requirements within 20 floors of the truth.~~ If the distances are too large for the requirements ~~expected~~ of coming designs, another variant is produced, ~~chosen by the~~

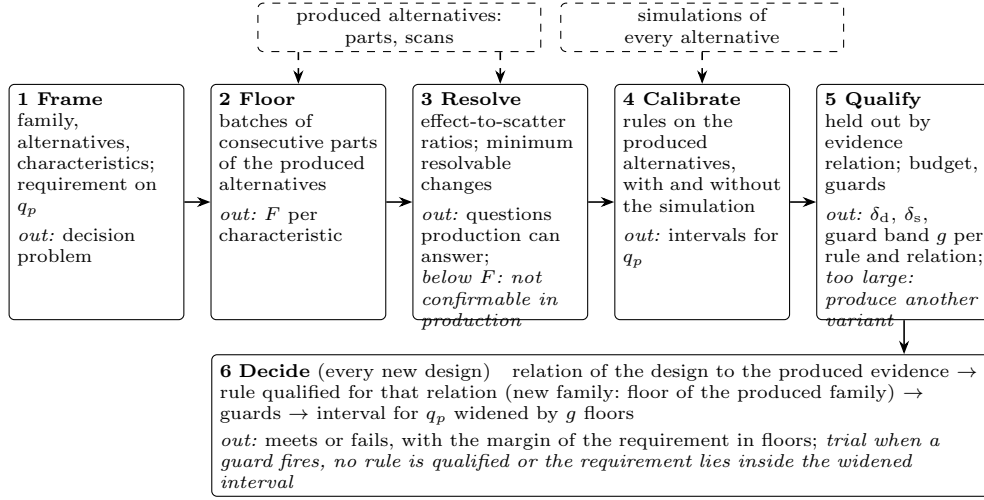


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

calibration design. Step 6 decides: it establishes the relation of the new design to the produced evidence, applies the rule qualified for that relation and the guard, and returns a verdict it and the guards, widens the rule's interval by g floors and returns meets or fails with the margin of the requirement in floors, or a trial when the guard fires or the margin lies below the rule's safe distance. A guard fires, no rule is qualified or the requirement lies inside the widened interval.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzlmann, 2026; Baum et al, 2026a). Square blanks of DP600 dual-phase steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (operation OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in the curvature of the base shape, the wall angle and the bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine oil application, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m² of oil), giving 18 alternatives, each produced as one series of 500 consecutive parts. Within a geometry the new designs are therefore new process settings and lubrication variants on an existing tool; the change of geometry is the only change of product design. For every part the dataset holds the press force, traverses of the sheet thickness and of the oil film before forming, press signals, sheet and oil-film traverses and 3D scans after drawing and after cutting (0.077 cutting; 10 of the 9,000 parts carry no scan. The series start at punch temperatures of 20.5–29.6 mm per pixel across the scan line, 0.158°C and warm up by 1.0–5.0 K over their first 150 parts; their median sheet thickness ranges

from 0.986 to 0.993 mm per line and 0.008, and the 100 kN series ran at a stroke speed of 47 mm in height). Of the 9,000 parts, 10 carry no scan, leaving 8,990 /s against 78–79 mm/s for the others (ESM Table S10). The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a): LS-DYNA models of the same process on a quarter of the part, with nodes about 1 mm apart, a scaled DP600 flow curve, Coulomb friction with a constant coefficient between blank and tools, and springback after both operations; the documentation does not state the yield criterion or the unloading model, which govern springback (Yoshida and Uemori, 2002), and the models do not represent the stroke speed. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm in six steps) and the friction coefficient (0.05–0.15 in eleven steps) at each geometry and blank-holder force. A part is matched to the simulation with the nearest sheet thickness and the friction coefficient that the dataset’s linear mapping assigns to its measured oil film, an assumption of the dataset’s authors on which Mc. M3 and every matched simulation depend. The nominal simulation of an alternative is the one at 0.99 mm, the grid value nearest the measured median thickness of 0.990 mm, and friction 0.10, the middle of the range. The simulated characteristics carry extraction noise from the 1 mm node spacing; it is quantified in Section 5.2.

4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, all referred to the plane of the cup bottom (Table ?? ESM Table S1): the mid-side and corner draw-in and the flange waviness, process indicators of material flow and wrinkling on the flange that the cut removes, and four characteristics that a drawing would carry: the wall angle (design angle 20° concave and 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on their absolute values. The four sides of a part measure several of them twice, and this redundancy exposes the measurement (Table ??). The flange width in the direction ESM Table S8): the flange width along the scan lines scatters 4 to 18 times more than across them, uncorrelated with it and without serial correlation, and reads 2.3–3.5 mm larger; so the mid-side draw-in is therefore taken from the widths taken across the scan lines only. The, and the north wall carries about 0.3° of uncorrelated scatter in every series and reads 0.9–1.6° low on the convex cups, uncorrelated scatter and the south wall is shadowed; so the wall angle is therefore the mean of the east and west walls, which also cancels a residual tilt of the part (on the convex cups the two are anti-correlated). The corner. The mid-side draw-in uses the mean of the two flange diagonals, which differ by 1.6–2.2 is quantised in steps of 0.019 mm in every series, consistent with a small non-orthogonality of the scan axes to which their mean is insensitive. The quarter-model simulations are symmetric about the diagonal, so their values along

~~both axes estimate the same quantity and are averaged. Differences that remain,~~
0.38 floors, by the pixel pitch; the other characteristics are effectively continuous.
~~Differences between scans and simulations—the simulations, such as the scanned~~
~~surface against the shell mid-surface, edge treatment and spatial resolution—affect~~
~~enter the absolute offsets between the two but not the floors and the effects within~~
~~production.~~

4.4 Evaluation design

Every produced alternative is judged as a new design under ~~four schemes of~~ grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight alternatives of its geometry calibrate, so two siblings at its blank-holder force ~~, produced with another lubrication pattern,~~ are among them. *New process setting*: ~~all the~~ three alternatives of one ~~blank-holder~~ force of a geometry are held out together and the six others calibrate; the 300 kN alternatives are ~~then~~ interpolated and the 100 and 500 kN alternatives extrapolated in force. *New family*: the alternatives of one geometry calibrate the rules for the other. ~~The pooled,~~ which gives two transfers. Pooled versions of the first two add the other geometry's alternatives to the calibration set and test whether the production record of another family helps. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M2/M2n, M3/~~M4~~M3n, M5/M5n), so that the paired difference isolates what the simulation contributes as it is used, and the nearest produced setting (~~NN~~) is the baseline without simulation and ~~without~~ learning. A scope comprises 126 cases (18 alternatives, 7 characteristics)~~and two transfer calibrations.~~

The distances are computed exactly~~from the cases.~~ Their intervals come from 1000 resamples of the held-out alternatives within each geometry, ~~every resample taking one each with a~~ block-bootstrap replicate of the true q_p of ~~each alternative (batches of 50 parts resampled), so that the uncertainty of the reference enters; the resamples are every alternative,~~ common to all rules ~~, which gives paired intervals for their differences. These so that their differences get paired intervals (ESM Table S13); these intervals hold the fitted rules fixed; a.~~ A second bootstrap resamples ~~the alternatives of whole lubrication patterns within~~ each geometry, which keeps every case in its evidence relation, and refits every alternative-level rule ~~on each resample and repeats the evaluation, which adds the variability of the calibration sets (ESM Section S1).~~

4.5 Computation and development record

The constants of the floor and the requirement grid were ~~chosen fixed~~ before the decision results were seen~~and are listed in Appendix ??;~~ as the repository history records~~their dates.~~ The rules were not~~all fixed in advance~~: M0–M5 ~~and the three original calibration scopes~~ were defined in a first analysis, ~~the conformalised Gaussian process M6 was added as a candidate during it, and the rules;~~ M0w, Mc, M1n, NN, M2n~~and~~, M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes ~~and the exact estimator of the distances were added in a second round of analysis, after the results of the first~~ were added after the first results had been seen. ~~Every variant tried is reported, and;~~ and a review of that version added M1s, the

jackknife+ intervals, the ~~comparisons between rules carry~~ between-run noise floor and the analyses of Sections 5.4 and 5.7. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or the ESM. The pipeline reads single files from the public archives ~~by HTTP range requests, keeps only the extracted features,~~ and reproduces every result from the cached feature tables in about four hours on four processor cores.

5 Results

5.1 What the production floor contains

~~Table 3 gives the floor of each characteristic. Two~~ In 95% of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in, 0.046 mm in corner draw-in, and 0.078° in wall angle, 0.11° in arm angle, 0.011 mm in cup depth and 0.007 mm in bottom dome at 95% confidence; resampling whole alternatives instead of batches, which respects the dependence of neighbouring batches under drift, widens the intervals most for the waviness (0.0021–0.0050 mm) and the arm angle (0.089–0.15°) (Table 3), and one floor is 0.66 to 2.2 long-term standard deviations of the parts. The floors are set by drift ~~: once within the run; without~~ the time order of the parts ~~is removed,~~ they shrink by a factor of 2.1 (wall angle) to 4.5 (corner draw-in), and over a series the punch temperature rises by a median of 4.3 K (range, batches one position apart differ by 0.51–0.80 floors and nine positions apart by 1.1–6.4 1.5 floors, and the normal model of Section 3.2 reproduces the floors to within 12 K), one of the sources of this drift.

~~Table ?? shows what the floor contains. Part-to-part noise that is independent between parts—which % (ESM Tables S3 and S4). Independent part-to-part noise, which includes the repeatability of the scanner—would scanner, would produce 15–33% of each floor, so the floor is dominated by variation between batches, which a repeatable measurement cannot create. The redundancy of the scans had shown where the measurement did matter: with the flange widths along the scan lines and the north wall included, and the resolution of the measurement is small against the floor except for the mid-side draw-in had a floor of 0.085 mm and a part standard deviation of 0.13 mm, of which the redefinition removes 40% and 70%; the wall angle kept its floor (0.081 against 0.078°) because the north wall’s scatter averaged out within batches. Drift (ESM Tables S2 and S9); assumption A3 holds for repeatability and resolution, but the stability of the scanner itself cannot be separated from drift of the process cannot be checked without a reference artefact scanned through the series. The floor also depends on the protocol: from the first 250 parts of every series it is 0.49–0.95 of the full value, and it varies by a factor of 0.6–2.2 over batch sizes of 25–100 and quantiles of 0.90–0.99 (Section 5.9). Nor do the process signals identify the source of the drift: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20 , and removing the predicted drift changes the floors by -16 to $+12\%$ (ESM Table S30). Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across these series is 1.5 to 9.1~~

Table 3 Production floor per characteristic, pooled and per geometry

Characteristic	F	95 % interval		Concave	Convex	F/σ_{LT}	Drift	Step
		batches	series					
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.042–0.061	0.050	0.052	1.2/1.5	3.3	0.38
Draw-in, corner (mm)	0.046	0.035–0.052	0.032–0.053	0.053	0.030	1.7/1.3	4.5	<0.01
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0021–0.0050	0.0037	0.0021	2.1/0.8	2.5	<0.01
Wall angle (°)	0.078	0.066–0.10	0.054–0.10	0.081	0.078	1.6/0.7	2.1	<0.01
Arm angle, cut (°)	0.11	0.094–0.12	0.089–0.15	0.12	0.100	1.9/0.9	2.6	<0.01
Cup depth (mm)	0.011	0.0096–0.012	0.0090–0.012	0.0087	0.012	2.2/1.7	3.9	<0.01
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0052–0.0084	0.0034	0.0082	1.4/1.5	3.9	<0.01

Batches of 50 consecutive parts, 95 % quantile of the difference of batch centres. Intervals: 2000 resamples of batches within series, and of whole series. F/σ_{LT} : floor over the median long-term standard deviation of the parts (concave/convex). Drift: floor over the floor of randomly permuted series. Step: measurement resolution in floors.

times the reference. The decided statistic itself reproduces to about one floor: the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors at-in 95 % confidence, and the block-bootstrap standard error of an alternative's q_{95} is 0.11 floors in the median (at most 0.39) of pairs. Distances below about one floor are therefore at the limit of what production can confirm. Variation between runs is not in the floor: the three lubrication series of one geometry and force, which production mostly cannot tell apart, differ by a median of 0.5–1.6 floors and, at 95 %, by 1.2 (waviness) to 9.8 floors (dome), an upper bound of what separate runs of one design add beyond what production, or a trial, can confirm.

5.2 What production resolves and how far the simulation misses it

Figure ?? shows the 18 alternatives against the floor and Table ?? counts the contrasts that production resolves. All 63 geometry contrasts exceed the floor, with median effect-to-scatter ratios from 2.8 (waviness) to 129 (wall angle). Blank-holder and blank-holder force is resolved in 105 of 126 contrasts; the exception is the wall angle, which force barely moves (4 of 18). The lubrication pattern is resolved, but the lubrication pattern only in 51 of 126 contrasts and in at most 11 of 18 for any characteristic, so that for most characteristics an alternative's (42 with a bootstrap probability of at least 0.95), so a lubrication variant is often a near-replicate of it. The force sensitivity is local: between 100 and 300 kN, where the stroke speed changes as well, the minimum resolvable its siblings (ESM Table S5). The minimum resolvable force change is 21–67 kN for the cup depth, the waviness, the draw-ins and the arm angle, whereas between 300 and 500 kN, at equal speed, it is 43–105 between 100 and 300 kN for the depth, the draw-ins and the waviness and the arm angle no longer responds; most characteristics and lies beyond either force interval for the wall angle and the dome need 366–731 kN in either interval.

5.3 Simulation against production

875 The nominal simulation misses production by far more than a floor (Table ??). Its:
876 its median offset reaches 109 floors for the cup depth of the concave cup and 108 for
877 the corner draw-in of the convex cup, and within one geometry the offset varies by
878 8.5 to 39 floors between alternatives. In units, the simulation overestimates the corner
879 draw-in by about 3 mm and the mid-side draw-in by 0.6–1.2 mm and underestimates
880 the wall angle by 1.0–1.2°. Its, its blank-holder force effect on the draw-in is 2.6
881 2.7 to 5.1 times the measured one, which points to the friction model—a constant
882 Coulomb coefficient where friction falls with contact pressure (Hol et al, 2012)—and
883 to the modelled blank-holder contact rather than to the solver. The signed comparison
884 exposes a sign error that the decision scale hides: and the cut arms of the concave cups
885 bend upwards in every part and downwards in 89–100 % of the simulations, so their
886 absolute values agree better than their physics. The offset of the cup depth,—0.90
887 to —0.95 but upwards in every part (ESM Tables S6 and S7). Part of the offset is
888 definitional: the cup depth differs by a median 0.90 and 0.95 mm for all alternatives, is
889 nearly constant and is consistent with a difference of reference—the shell mid-surface
890 against in the two geometries, consistent with the scanned surface, and the drawing
891 depth of the model—rather than with forming physics. The simulated characteristics
892 also carry extraction noise from the 1 mm node spacing (Table ??): 1.7–3.1 floors
893 for the wall angle and 2.4–9.8 floors for the dome, which inflates the upper end of
894 the simulation envelope of the dome by up to 21 against the shell mid-surface, and
895 removing this offset alone brings the decisive distance of the nominal simulation from
896 108 to 65 floors.

897

898 5.3 Decisions by evidence relation

899

900 Table 4 and Fig. 2 score the ~~twelve~~ rules. The nominal simulation (M0) needs require-
901 ments 108 floors from the truth (bootstrap interval 101–115) before 95 % of its verdicts
902 are right, ~~and~~; it errs mostly by rejecting adequate alternatives: ~~its safe distance is,~~
903 with a safe distance of 17 floors for false accepts but 109 for false rejects (Table ??).
904 Reading the process-window simulation by its worst case (M0w) makes it neither safe
905 nor decisive (94 and 119 floors), because the window does not contain the simulation's
906 bias.

907

908 *A new variant with a produced sibling.* With the other eight alternatives of the
909 family produced, every calibrated rule improves ~~by an order of magnitude on the~~
910 nominal simulation, and the simplest ~~one~~ is the most decisive: the nearest produced
911 setting (NN), the mean q_{95} of the two siblings at the same ~~blank-holder~~ force, is right
912 beyond 3.4 floors (2.0–4.5) ~~and at 10 floors decides every case correctly. The Gaussian~~
913 ~~process without the simulation (M5n) decides at 6.1 floors and is safe at 1.0, and with~~
914 ~~the simulation (M5) at 6.7 and 1.4. The interval rules are safe at or below 1.4 floors,~~
915 but the envelope with the residual margin (M2)—, 0.17 mm of mid-side draw-in. The
916 scaled simulation follows at 4.9 floors, the Gaussian processes with and without the
917 simulation decide at 7.2 floors and are safe below one floor, and the envelope rule is
918 safe at 0.15 floors but sends a third of the designs to trial at 10 floors and ~~is decisive~~
919 decides only at 23 floors. Bias-correcting the nominal simulation (M1, 15 floors) is
920 worse than taking the median of the produced alternatives without it (M1n, 9.1), ~~and~~
~~calibrating the friction coefficient (Mc, 27 floors) is worse still.~~

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

Rule	New variant		New process setting				New family	
	δ_d/δ_s	FA	δ_d/δ_s	FA	interpolated	extrapolated	δ_d/δ_s	FA
M0	108	17	108	17	109	106	108	17
M1	15	15	24	24	11	25	35	33
M1s	4.9	4.7	19	164	19	19	122	122
interval	4.7–6.3		14–21		19–20	15–19	121–123	
M2	23/0.15	0.30	18/8.1	8.3	18/2.7	18/9.9	51/16	11
M5	7.2/0.75	0.15	19/6.8	7.6	19/2.8	19/8.9	38/28	32
interval	6.6–8.4		18–20		18–20	18–20	35–39	
M5n	7.2/0.55	0.10	17/6.1	9.0	16/0	18/8.6	132/130	167
interval	6.4–7.5		16–18		15–16	16–19	130–133	
NN	3.4	2.8	8.5	10	4.7	9.5	131	169
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131	

Over all seven characteristics; one number for rules that always decide. FA: safe distance on the false-accept side (failing alternatives). Interval: bootstrap 95 % interval of the decisive distance (fits fixed). 126 cases per relation (42 interpolated, 84 extrapolated); a new family rests on two transfers. All rules: ESM Table S12.

A *new process setting*. When every alternative at the new design’s ~~blank-holder~~ force is held out, all rules lose. The nearest produced setting is right beyond 8.5 floors (5.5–10.2) ~~and the median of the produced alternatives beyond 10~~; the interval rules ~~are decisive only at 13–18 floors and safe at 5.4–8.1 (M4 13/7.9, M2n 15/decide only at 13–31 floors and are safe at 5.9, M2 18/8.1, M5 18/7.4, M5n 18/5.4; M3 32/16)–16~~, and at 10 floors they send ~~a fifth to a third 18–37 %~~ of the designs to trial. Interpolating in force (300 kN from 100 and 500 kN) is much easier than extrapolating: the nearest setting decides at 4.7 against 9.5 floors, and M2 and M5 are safe at 2.7 and ~~2.9–2.8~~ against 9.9 and 8.9 floors. ~~The scaled simulation decides at 19 floors and, extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from their limit.~~

A *new design family*. ~~Calibrated only on the other geometry~~ In the two transfers, every rule without the simulation fails (131–133 floors), because nothing in one family’s production record describes the other. With the simulation, the bias correction decides at 35 floors, M5 at 38 and ~~the boosting model M3 at 4849~~, and the envelope ~~with its residual margin rule~~ is the safest ~~rule~~ at 16 floors; no rule is safe closer than that. ~~The simulation thus reduces the distance at which a new family can be decided from about 130 floors to 35–51, but not to a distance at~~ ~~Scored in the floor of the produced family~~, which a design team would ~~rely on it near the requirement have~~. M5 decides at 35 and is safe at 31 floors and the envelope rule at 41 and 26. Sixteen floors are about 0.8 mm of mid-side draw-in or 1.3° of wall angle; ESM Tables S17 and S18 give the distances per characteristic in floors and in units.

5.4 Where the distances come from

Three features of the evaluation design shape the distances (ESM Table S21). First, the sibling relation mixes replication with variation. In 58 of the 126 new-variant cases neither sibling differs resolvably from the held-out alternative, and the nearest

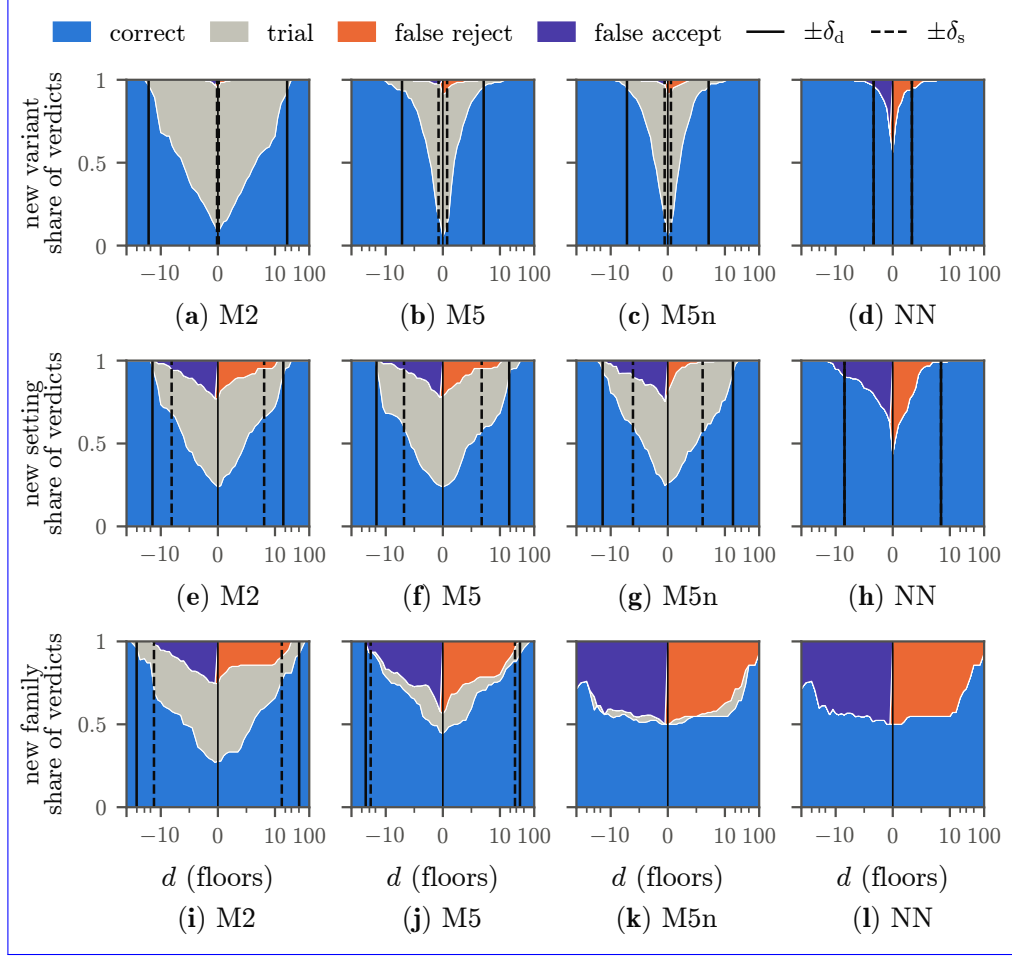


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ± 10 floors; 126 cases per panel) with the decisive and safe distances marked

produced setting is then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs; when both siblings differ resolvably (34 cases), it needs 4.9 floors and M5 9.7. Second, the extrapolation cases are confounded with stroke speed: the production-only rules lose almost only at 100 kN, where the press also ran slower (the nearest setting decides at 11 floors against 4.9 at 500 kN), whereas the simulation-based interval rules are unsafe at both ends. Third, the production record of the other family does not help within a family: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors for a new variant, M2 from 23 to 53, M2n from 19 to 325), although it raises the coverage of M5 from 86 to 90% and makes the centre of its interval the most accurate point prediction within the family (2.9 floors against 3.4 for the nearest setting).

The relation therefore does not govern decision fitness in general. Over the eleven calibrated rules and all three relations, the relation explains 69% of the variation of

the log decisive distance and the rule 6 %, but this share comes from the new family, where the rules without the simulation fail by construction; within a family the rule explains 68 % and the relation 15 %. The best attainable distance nonetheless grows with the novelty of the design, from 3.4 floors with a sibling to 8.5 for a new setting and 35 for a new family.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds (Table ??). Within a family as it is used (ESM Table S15). Added as an offset, it adds nothing within a family and degrades two rules: the decisive distance of M5 differs from equals that of M5n by +0.6 floors (95 % interval (difference 0.0 to +1.6), that floors, -0.4 to +1.1), and those of M2 from and M1 exceed those of M2n by +3.9 (+0.4 and M1n by 3.9 (+0.5 to +7.4) , and that of M1 from M1n by +6.2 and 6.2 floors (+4.6 to +9.7); for a new process setting the bias correction loses 13.5 floors by using the simulation, because the simulated force effect is too large. Rescaled, it adds information within a family, where the scaled simulation decides 4.2 floors closer than the median of the produced alternatives (-5.4 to -0.5) though 1.5 floors less close than the nearest produced setting, but for a new setting the rescaled trend is worse than none. Only for a new family is the simulation decisive, by about 93 floors. Learning adds little over the nearest produced setting: within the family M5 is 3.3 floors less decisive than NN (+2.1 to +5.1) and 2 floors safer, and for a new setting 9.4 floors less decisive. The ranking between learners rests on their 94 floors, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own design angle decides at 2.0 floors, against 7.3 for M5 (ESM Table S19). A surrogate of the simulations reproduces them to within 1.8 floors for 12 of the 14 characteristics and geometries, yet its decision fitness is that of the simulation it reproduces.

Learning adds intervals rather than on their predictions: used accuracy. Used as point rules, the centres of the M5, M5n and M4-M3n intervals decide at 3.4, 2.8 and 4.8 3.3, 3.1 and 3.7 floors within the family, close to as close as the nearest produced setting (Table ??). The part-level models keep their ranking under other learners and settings, including quantile-loss boosting and settings selected by an inner leave-one-out (Table ??).

The intervals are calibrated only where the calibration alternatives are exchangeable with the new design (Table ??). Within the family the residual margin of M2 covers the true q_{95} in 88.9 % of the cases—exactly the $n/(n+1) = 8/9$ that eight residuals attain—and the part-level jackknife of M3 in 93 %; , so the ranking of the learned rules rests on the width of their intervals (ESM Table S16). With the noise floor set by the variation between runs, the Gaussian-process intervals cover 71–72 85–86 % although their nominal level is of the true q_{95} within the family against a nominal 90 %, and they are narrow (1.1–1.2 floors in half-width) because the fits interpolate: the noise level sits at its lower bound in 35–38 the largest residual margin 89 % of the fits and a length scale at a bound in 28–48 (it guarantees at least $(n-1)/(n+1) = 7/9$ under exchangeability, which these cases lack) and the jackknife+ interval of M3 96 % (Table ??). For ; for a new setting every calibrated

interval covers only ~~39–56~~43–55 %, and for a new family the Gaussian processes cover none, M2 47 % and M3 40 at most 12 %. What a designer observes for a single verdict is the margin between the predicted interval and The inductive biases explain much of this: trees do not extrapolate, two force levels identify at most a linear trend, and in the median new-setting fit the length scale of the requirement. With requirements spread over the 20 floors around the truth, the share of correct verdicts among the decided ones grows with that margin for a new variant and a new setting—for the nearest produced setting from 86 % at any margin to 95 % beyond 5 floors and 99 % beyond 10 for a new setting—but not for a new family, where M5 is right in 72–75 % of its decided verdicts whatever the margin. The observed margin is a guide to a verdict only where the rule is calibrated for the evidence relation force sits at its lower bound, so the Gaussian process reverts to its mean (ESM Tables S23 and S22).

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1072 5.6 Calibration budget ~~and calibration design~~

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Figure 3 and ~~Table ??~~ ESM Table S20 show how the distances depend on the number of produced alternatives of the family and on how the new design relates to them. A single produced sibling decides: with one alternative at the new design's force, the bias correction, the ~~residual-margin envelope~~ rule and the nearest setting ~~all coincide by construction (the simulation terms cancel and a single residual gives no margin)~~ and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors, ~~and further~~. Further alternatives do not improve the nearest setting. ~~They~~; they make the bias correction worse (12–18 floors), ~~because the offsets of other forces dilute the sibling's~~ the envelope rule safer but less decisive, and the ~~interval rules safer but not more decisive~~ : M2 ~~becomes safe at 0.2–2.2 floors and decisive at 15–23~~ Gaussian processes safer and more decisive (from 12–14 floors with two alternatives to 7.2 with eight). Without a sibling, bracketing the new force buys safety but not decisiveness: ~~with produced alternatives on both sides, M2, M5 and M5n from two alternatives on, the interval rules are safe at 0.5–4.0–4.8 floors ; M2 and M5n are never wrong with no wrong verdict at 10 floors and M5 in under 0.5 %; but they~~, but decide only at 15–25 floors, ~~while the nearest setting decides at against 4.7–6.0 floors for the nearest setting~~. Extrapolation stays unsafe at every budget: ~~no rule is safe closer than 8.5 floors, and at 10 floors 2–13 % of the verdicts of the interval rules and the nearest setting are wrong, and 26–28 % of those of the bias correction. Averaged, and averaged~~ over all subsets ; the distances shrink with the number of produced alternatives mainly because the share of subsets that contain a sibling grows, from 46 % ~~of the cases~~ with two alternatives to 96 % with six; ~~what a produced alternative is worth depends on its relation to the designs to come.~~ . The odd-even pattern of the bias correction in Fig. 3 comes from its median offset, which averages the offsets of two forces only for an even number of alternatives. With calibration series of 50, 100 or 250 parts instead of 500, no distance within a family moves by more than 1.2 floors, and for a new family by at most 4 floors (ESM Table S29): the qualification needs short runs of several variants, and only the floor needs a run long enough to show the drift.

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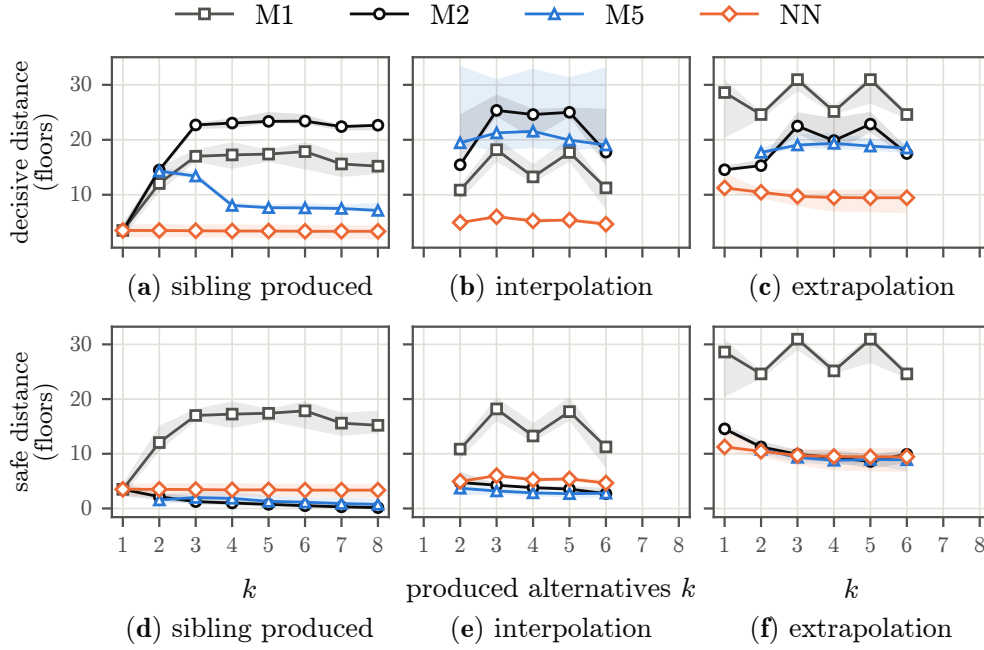


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95 % intervals over the held-out alternatives

5.7 Applicability guard

5.7 The procedure's decision rule

A guard is useful only where it can fire, which within this design means a new process setting and a new family (Table ??). For a new setting the range guard refuses exactly the extrapolated forces, the 84 of 126 cases whose distances are largest, and among the interpolated cases it lets through, M2. What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): for a new variant at least 95 % of the decided verdicts of the interval rules are correct at every margin; for a new setting the share grows with the margin, for the nearest setting from 85 % at any margin to 95 % beyond 5 floors; for a new family M5 and is right in 71–74 % of its decided verdicts whatever the margin. Step 5 turns this into guard bands (ESM Table S24): none for the interval rules within the family and 0.25 floors for the nearest setting; 2.0 floors for an interpolated setting, with which the nearest setting are never wrong decides at 6.7 floors, is safe at 2.7 and sends no design to trial at 10 floors. The novelty guard, which compares the simulated q_{95} of the new design with the calibration range, also refuses 84 cases but not the same ones; among the 80 verdicts it lets through, 7.0 floors for an extrapolated one, with which it decides at 17 floors and sends a quarter of the designs to trial at 10 floors; and for a new family no guard band up to 20 floors reaches 95 %, so the procedure sends every design to trial. The guard bands are chosen on the evaluation data, which makes the trial rates optimistic.

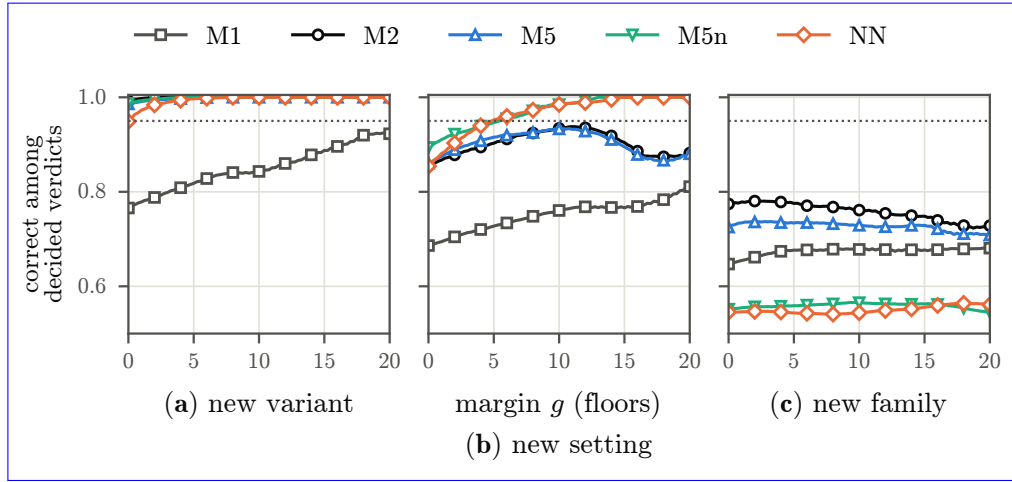


Fig. 4 Share of correct verdicts among the decided ones against the observable margin between interval and requirement, for requirements within 20 floors of the truth

The applicability guards act on the same relations (ESM Table S25): for a new setting the range guard refuses exactly the extrapolated forces, after which M2 and M5 are wrong twice and the nearest setting never, but 72–96% of the verdicts it refuses make no wrong verdict at 10 floors, whereas the novelty guard also refuses many cases that would have been correct. For a new family the family and range guards refuse everything. The novelty guard lets 12 of 126 cases through, all of them decided correctly.

A worked example. A tryout engineer has to decide whether the concave cup at 300 kN with medium lubrication, a new process setting, can be released without a trial run, with the six alternatives at 100 and 500 kN produced and a requirement that 95% of the parts have a mid-side draw-in or waviness, and among their 24 verdicts at 10 floors M5 is wrong in 6 (25% of at most 15.30%) and M1 in 6, whereas mm. The floor is 0.050 mm, the relation is interpolation and no guard fires. The nominal simulation predicts 15.66 mm and fails the design. The nearest produced setting predicts 14.99 mm, a margin of 6.2 floors, more than its guard band of 2.0 floors, so step 6 returns meets; beyond a margin of 6 floors every decided verdict of this rule for an interpolated setting was correct. M2 is never wrong and sends 12. M5 and M5n return intervals reaching 15.37, 15.58 and 15.33 mm and send the design to trial. A guard on the simulated value therefore protects a rule only as far as the rule's own interval does. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the limit: the procedure was right, the simulation wrong and the interval rules cautious.

5.8 Requirements from general anchored in capability and in tolerances

Requirements from the angular classes of ISO 2768-1 (ISO, 1989), applied to the wall angle against its design angle and to the arm angle against flat with the row for the shorter leg of each, turn the verdicts into concrete design decisions (Table ??). Of the

108 decisions, 69 concern an alternative that fails its limit, and the limit lies within 5 floors of the truth in 32 and within 10 floors in 55; deciding the worst side of a part instead of the mean of its sides would change the truth in 12 of them, since a general tolerance applies to every feature. For a new variant Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile, inside every decisive distance (ESM Table S26). Every alternative meets them, so a verdict is right, a false reject or a trial. At a performance index of 1.33 the nearest produced setting decides all 108 correctly and M5 still says meets in 91 % , with no false accept; within 5 floors of the limit, of the new-variant cases, because most of its errors are far smaller than its 95 % distance, whereas M5 is right in 75% says so in 44 % and sends the rest to trial, M2 is right in 25 %, and the bias-corrected simulation accepts three of the six failing alternatives there. For a new process setting the nearest setting is right in 97 % and the Gaussian process without the simulation in 92; for a new setting the two are right in 65 and 32 %, whereas M5 and M2 send 22-25 % to trial and falsely accept 4 % of the failing alternatives. For for a new family M5 is right in 68 % but rejects 46% no rule exceeds 54 % of the adequate alternatives and accepts 9 % of the failing ones; M2 never accepts a failing alternative but sends 39 % to trial. With a false reject costing half a false accept, the cheapest rule per decision is the nearest produced setting. Release-level requirements are therefore decided at the design stage only within a family, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario that mostly tests whether a rule recognises gross failures: only 32 of its 108 decisions lie within 5 floors of the limit (ESM Table S27). Its cheapest rule depends on the costs (ESM Table S28); for a new variant at any trial cost, the interval rules without the simulation (M2nthe nearest setting at any cost, M4 and M5n, tied) for a new setting the production-only interval rules while a trial costs at most a tenth of a false accept and the nearest setting from a fifth beyond, and for a new family M2 up to a tenth and a trial of every design while a trial costs at most a tenth of a false accept, the envelope rule at a fifth and M5 from a fifth. A drawing of a deep-drawn part would normally carry profile and position tolerances to datums (ISO, 2017a, 2021); which the data do not allow to evaluate; the general tolerances serve as requirements with realistic margins beyond, for a false reject at half the cost of a false accept.

5.9 Robustness

Table ?? repeats the analysis The analysis was repeated under variants of the data, the constants and the Gaussian-process specification. The variants rescale the distances but leave the conclusions in place: in every variant that includes it, the nearest produced setting remains the most decisive rule for a new variant and a new setting and M2 the safest for a new family, and the rules without the simulation fail for a new family (91 floors or more); only rules within a few floors of each other change places, such as M1, M5 and Mc for a new family. Correcting every characteristic for the model specifications (ESM Table S29). Adjusting for punch temperature changes the pooled floors by —14 to +4 % and the distances of the calibrated rules by at most no calibrated distance by more than 2.6 floors. Smoothing the simulation removes its extraction noise but not its bias: M5 is unchanged within the family and the envelope

~~rule becomes less safe~~ Dropping the warm-up of the first 150 parts shrinks the floors
 and so lengthens the distances (the nearest setting 4.1 floors with a sibling, 12 for a
 new family (25 instead of 16 floors). Deciding on the 90th or 99th percentile instead
 of the 95th moves no distance by more than 1.7 floors. Floors from batches of 25
 or 100 parts rescale the distances by factors of 0.80–0.96 and 0.97–1.16 (setting), and
 floors at the 90 % or 99 % quantile by 1.16–1.46 and 0.54–0.83. Requiring 90 % or 99 %
 correct verdicts instead of 95 % shortens the decisive distances by up to 43 % (within
 the family) or lengthens them by up to 93 % (for a new setting), as an operating
 characteristic should. The taken between series, 1.5 to 9.1 times the reference, shorten
 them (0.75 and 2.7 floors). Other floor protocols rescale the distances by 0.60 to 1.42,
 the Gaussian-process variants—a Matérn kernel, a linear trend, one-hot lubrication,
 a floor on the length scales and the first specification without the noise floor—keep
 M5 at 6.4–6.9 floors for a new variant and 17.7–18.9 for a new setting. Leaving out
 one lubrication pattern behaves like a new variant (nearest setting 3.4, M5 6.4/1.1
 floors), because the siblings at the same force remain. Refitting every rule on resampled
 calibration sets specifications move them by at most 2.1 floors, and in the unit of the
 reproducibility of g_{95} the nearest setting decides at 2.8 and 6.9 floors. In every variant
 that includes it, the nearest produced setting remains the most decisive rule within
 a family and the envelope rule the safest for a new family. The refitting bootstrap
 widens the intervals, most for the Gaussian processes of the learned rules most (M5
 4.0–11 floors for a new variant; 5.0–18.6 floors, against 2.2–7.0 for the nearest setting
), whose fits on calibration sets with repeated alternatives are unstable. Replacing
 each part's oil film by its pattern's median makes M3 less decisive (33 instead of
 17 floors), and a conformalised Gaussian process (M6), the 2.3–4.6, yet the nearest
 setting is the more decisive in all 200 resamples, against M5 mean with a margin on
 standardised leave-one-out residuals, is safe at 0.4 floors within the family but decisive
 only at 39: it buys safety for the over-confident Gaussian-process intervals at the cost
 of decisiveness and M5n and for a new setting alike (ESM Table S15).

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1272 6 Discussion

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1274 6.1 What the evaluation establishes

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Five findings answer the ~~three research questions~~ beyond the particular
~~numbers~~ research questions. Two follow from the definitions and hold wherever the
 framework is applied. First (RQ1), ~~production has a resolution, and the floor makes~~
~~decision fitness comparable across characteristics and predictors: it is set by drift~~
~~within a run rather than by the scatter of single parts or by the measurement, the~~
~~decided percentile reproduces to about one floor, and the distances of very different~~
~~rules can be read on one scale. It the floor puts every rule and characteristic on a~~
 common production-referenced scale, but not on a common scale of nonconforming
 fractions, and as a within-run quantity it is a lower bound for series production, since
 separate runs of one design may differ by several floors more. Second of the variation
 of series production. Second, an interval rule trades decisiveness for safety, the two
 distances make the trade explicit, and the nominal level of an interval is reached
 only where the calibration alternatives are exchangeable with the new design. Three

are observations on this dataset and hypotheses for confirmation. Third (RQ2), the relation of a new design to the produced evidence governs decision fitness more than the model does. With a produced sibling, the best rules calibrated on production decide within a few floors; best attainable distance grows with the novelty of the design, from 3.4 floors with a sibling (0.85 when the sibling is a near-replicate) to 8.5 for a new process setting no rule decides closer than 8.5 floors, and extrapolating in force doubles the decisive distance of the nearest produced setting and more than triples the safe distances of M2 and M5; for a new family no rule is safe closer than 16 floors. Third setting (4.7 interpolated, 9.5 extrapolated with a speed change) and to no rule safe closer than 16 floors in two transfers, while within a family the choice of rule matters more than the relation. Fourth, production evidence buys safety before decisiveness: interval rules become safe with few produced alternatives that bracket the new design but remain indecisive, and, and decisive only with near-replicates make them decisive (Section 5.6). Fourth, interval rules trade abstention for safety, the two distances make the trade explicit, and the nominal level of an interval is reached only where the calibration alternatives are exchangeable with the new design. Fifth (RQ3), in this data regime the simulation's contribution is confined to a new family small-data regime and with this simulator, the simulation contributed to decisions only across families as an offset, a fitted scale made it useful within a family but harmful for a new setting, and learned models add calibrated abstention rather than accuracy over were as accurate as the nearest produced setting as point predictors while adding abstention.

6.2 Using the framework in design

Table 5 separates guidance that follows from the framework and plausibly holds elsewhere from observations on this dataset, with their evidence base. The general guidance turns the distances into planning questions basis. Before a requirement is fixed, the expected margin between the requirement and the capability of the design requirement and capability can be compared with the safe distance of the best rule qualified for the evidence relation of that relation of the design; if the margin is smaller, no design-stage decision can be trusted relied on, and a trial or further production evidence has to be planned. Release-level requirements make this the usual case: they lie within a few floors of the capability, where only a near-replicate production record decides reliably, which is consistent with the practice of requiring a production trial run for a process change. For a single verdict, design, step 6 turns the qualification into a verdict whose guard band a designer can see, and the margin between the predicted interval and the requirement is what a designer observes, and it is informative only where the rule is calibrated for the relation (Section 5.5) worked example shows that the most decisive rule is also the most transparent: the nearest produced setting points to the produced alternative that it copies, whereas a Gaussian-process interval does not. Which rule to prefer then depends on the costs: with the realistic margins of general tolerances, a rule that always decides was cheapest where it was accurate, and an interval rule where a false accept is expensive relative to a trial, and those differ by relation: a trial of a new force on an existing tool is a short press run, whereas a trial of a new family may need a new tool. In a company, steps 1 to 5 fit the start of

1335 **Table 5** Guidance for design-stage manufacturability decisions, with its basis

1336 Situation	1337 Guidance	1338 Basis
1339 <i>From the framework (general)</i>		
1340 Before trusting a predictor	1341 Measure the floor on consecutive production of the family with a stated protocol and report σ_b , σ_w and its lag dependence; treat it as a lower bound of series variation.	1342 definitions
1343 Qualifying a rule	1344 Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	1345 definitions
1346 Choosing a rule	1347 Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	1348 definitions
1349 Deciding a single design	1350 Apply the rule's guard band to the observable margin; a margin inside the guard band, or a fired guard, means a trial.	1351 definitions
1352 Fixing a requirement	1353 If the expected margin to the capability is below the safe distance of the best qualified rule, plan a trial; no decision is reliable closer than a trial reproduces the quantile.	1354 definitions
1355 <i>Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)</i>		
1356 New variant with a produced sibling	1357 nearest produced setting right beyond 3.4 floors (0.85 for a near-replicate); calibrated interval rules safe below one floor	1358 126 cases
1359 New process setting	1360 nearest produced setting right beyond 4.7 floors interpolated and 9.5 extrapolated; interval rules safe at 5.9–16 floors	1361 42 + 84 cases
1362 New family	1363 best rule safe at 16 floors; rules without the simulation fail; a transferable drawing nominal helps	1364 two transfers
1365 Release-level requirements	1366 lie within 1–3 floors of the 95th percentile, inside every decisive distance	1367 126 cases

1363 production or the production part approval of a family, when its series are measured
1364 anyway, and step 6 the later decisions on variants and process changes.

1366 6.3 ~~Where simulation and machine learning help~~

1368 6.3 Simulation and machine learning in this regime

1370 ~~The question the call for this collection poses about learning from manufacturing data~~
1371 ~~receives a sober answer here.~~ Within a design family , the production record carries
1372 ~~the decision: the nearest produced setting is the most decisive rule~~carried the decision,
1373 and adding the simulation ~~to as an offset did not improve~~ a calibrated rule~~does not~~
1374 ~~improve it and, for the bias correction, makes it worse,~~ because the ~~simulation's process~~
1375 ~~trends—the effect of blank-holder force above all—are not credible. A physically~~
1376 ~~motivated calibration of simulated force trend was 2.7 to 5.1 times too large. Rescaling~~
1377 the simulation repaired this within the family but extrapolated badly for a new setting,
1378 and calibrating the friction coefficient did not repair ~~this~~it, which points to the fric-
1379 tion law and the blank-holder model rather than to a parameter. ~~Learning models, a~~
1380 ~~Gaussian process or gradient boosting with or without the simulation, predict about~~

~~Learned models predicted~~ as well as the nearest produced setting; what they ~~add-is~~
~~an interval~~, ~~which makes a rule~~ added was an interval that is safe where the cali-
 bration alternatives are exchangeable with the new design and over-confident where
 they are not. ~~The simulation is indispensable only for a new family~~, ~~where nothing~~
~~in the production record of another family carries over~~, but even there it supports
 decisions only at margins that a design team would not need a model for. ~~A model's~~
~~interval is a statement about the model; the safe distance for the evidence relation~~
~~is a statement about production~~ A multi-task model cannot help a family with no
produced alternative, because the correlation between families cannot be estimated
without its data; the pooled Gaussian process, which carries the geometry as an input,
is the borrowing of strength that the data allow, and it improved coverage but not
decisiveness. These findings concern one simulator of low fidelity for this production,
used through offsets and one scale, and learners with six to nine produced settings
each; a surrogate of these simulations inherits the simulator's distance, whatever its
accuracy against the simulation. For the evaluation of AI-based DFM support ~~this~~
~~means that accuracy on the lesson is methodological~~: held-out ~~data accuracy~~ is not
 enough:-, ~~and~~ a learned model should be compared with a production-record base-
line~~and~~, scored by evidence relation -, in a unit that production defines, and shown
to its user with a guard band. Whether that makes designers' reliance match a rule's
reliability (Lee and See, 2004), and when an abstention should go to an expert rather
than to a trial (Madras et al, 2018), are hypotheses for a study with design teams.

6.4 Implications for design research

The framework gives ~~three several~~ constructs of design research a production-
 referenced measure. ~~The safe distance of a rule for an evidence relation is the~~
~~margin a design team has to keep between a prediction and a requirement for~~
~~the verdict to be trusted, and so puts a number on what design margins are for~~
~~(Eckert et al, 2019; Brahma and Wynn, 2020).~~ ~~The two distances decide~~ For variant,
adaptive and platform design (Pahl et al, 2007; Simpson et al, 2006), the distances
by evidence relation quantify how far production knowledge carries from produced
variants to a new one, and the nearest produced setting sets the baseline that any
capability database (Thornton, 2004) or learned model has to beat. For margins, the
safe distance is the part of a margin that covers predictive uncertainty, as distinct
from the part that absorbs change (Eckert and Isaksson, 2017; Eckert et al, 2019),
and the guard band is its counterpart for a single decision. For testing and prototyping
strategies (Thomke and Bell, 2001; Salado and Kannan, 2018; Camburn et al, 2017)
, the distances and the reproducibility of a trial say when a test is
cheaper than a decision, ~~the trade at the core of testing strategies~~
~~(Thomke and Bell, 2001; Salado and Kannan, 2018)~~ and how close to a
requirement a test can itself decide. For overlapped development
(Krishnan et al, 1997; Terwiesch et al, 2002) and set-based design (Sobek et al, 1999)
, they state when design-stage information is fit to act on and how wide a feasibility
limit has to be. And for verification and validation, the evidence relation is the
question of the application domain (Oberkampff and Roy, 2010), and the ~~calibration~~
~~budget turns the production of variants into a planned verification activity. And they~~

show how far simulation front-loads a decision (Thomke and Fujimoto, 2000): only as far as its distance for the relation at hand distances could serve as the evidence of a risk-informed credibility assessment (ASME, 2018). In the terms of the design research methodology (Blessing and Chakrabarti, 2009), the evaluation reported here establishes that the framework discriminates between rules and evidence situations on real production data—its empirical performance is a support evaluation: it shows structural validity in the sense of the validation square (Seepersad et al, 2006)—while its use by design teams, and the success criteria that would link the distances to fewer trials or late changes, remain to be evaluated, that the metrics discriminate between rules and relations on real data (Seepersad et al, 2006), and is a first step towards performance validity, which would require showing that decisions taken with the framework are better than decisions taken without it.

6.5 Applying the framework elsewhere Reuse for other processes and AI-based DFM tools

The framework needs four things: several produced variants of a design family, each with whose relations to each other span those of the coming designs, a series of consecutive parts or replicated batches of one design from which the floor can be measured; variants whose relations to each other span the relations of the coming designs; a per variant, a design-stage predictor for every variant, which may be a simulation, a surrogate or the production record itself; and characteristics that can be measured identically on parts and predictions. None is specific to forming predictor for every variant, and characteristics measured identically on parts and predictions. On this dataset the long series served the floor, whereas calibration runs of about 50 parts per variant qualified the rules as well as full series did; how many long runs a floor needs was not tested. Table A2 maps these onto the processes named most often in DFM: attribute outcomes such as splits, porosity or short shots call for a requirement on a rate rather than on a quantile, which the framework does not yet cover. A predictor need not return an interval: any tool that returns meets, fails or trial for a stated requirement, a classifier at its operating threshold or a language-model check given the requirement value, can be scored by sweeping the requirement. For continuous design spaces, the relation becomes a distance from the produced designs in descriptor space, or the position inside or outside their hull. A minimal report of such an evaluation comprises the floor with its protocol and variance components, the decisive and safe distances by relation with refitting intervals and one-sided values, the coverage of interval rules, the nearest-produced-setting and no-simulation baselines, and the guard bands and guard performance. Distances per characteristic rest on as many cases as there are produced alternatives, so a team would pool characteristics with similar ratios of floor to part scatter, or accumulate evaluations over families, before relying on them. An attempt to apply it the framework to an open dataset of powder bed fusion of PA12 (Leirimo, 2021) showed what the first condition demands: grouping the failed on the second requirement: grouping specimens by orientation class would have mixed several mixed orientations in a batch, and only eight identical anchor specimens per build replicate one design. Two-sided requirements follow from the absolute deviation from the target, as for the arm angle and the dome; requirements on a high

conformance share are scored in the same way as long as the series estimates the quantile, and need a parametric tail beyond it were too few for a floor; a between-build floor from replicated anchors would be the remedy.

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, and the relations within a family come from a 3×3 factorial in which one factor, lubrication, is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two ~~calibration~~ transfers between geometries that differ in three features at once, and the intervals of its distances hold the two fits fixed. Each alternative was produced once, so the floor covers drift within a run but not between days, coils or tool states; ~~the quasi-replicate series bound that variation from above~~, the reference q_{95} is that of one run, and the order and dates of the series are unknown; between-series variation bounds the missing component at 1.2–9.8 floors : No gauge study or acceptance test of the scanner (VDI/VDE, 2012; ?) was available: the redundancy of the scans bounds the independent measurement noise, but a (Section 5.9). A drift of the scanner cannot be separated from that of the process, so the floor is a production-and-measurement floor, and scan and systematic scan errors and differences of reference surface reference-surface differences enter the absolute offsets of the simulation, though not the floors and effects. The simulated characteristics carry extraction noise of up to several floors, and the documentation of the simulation leaves the material model open. The blank-holder force of 100 kN ran at a slower stroke. Lubrication is a pattern whose oil amount varies between series. The extrapolation in force is confounded with stroke speed, which the simulation does not represent, so the extrapolation results describe a force and speed change together. The requirements are one-sided limits on the 95th percentile of a characteristic averaged over sides, the scenario covers two angles, and the reference q_{95} has a standard error of about 0.1 floor which is below the conformance of series release. Several rules, the grouped schemes and the estimator were added after the first guard bands were added or chosen after earlier results had been seen and are evaluated on the same data, so the comparison between rules carries selection optimism (Cawley and Talbot, 2010); the analysis was carried out by a single analyst, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production ,in a unit on a scale that production defines. The production floor—the difference that drift and scatter produce between two batches of one design—expresses floor expresses the fitness of any predictor for a decision independently of the predictor and the characteristic, and in the units of the characteristic; the decisive and safe distances, read as operating characteristics, tell how close to a requirement a rule decides and how close it can be trusted relied on not to mislead; and a guard band turns them into a rule for a single decision.

On 18 deep-drawn design-process alternatives with matched simulations, the relation of a new design to the produced evidence governed decision fitness more than

1519 ~~the model did. With best attainable distance grew with the novelty of the new design:~~
1520 ~~3.4 floors with a produced sibling~~~~the production record decided within a few floors,~~
1521 ~~8.5 for a new process setting~~~~no rule decided closer than 8.5 floors, and for a new family~~
1522 ~~no rule was,~~ ~~and no rule~~ safe closer than 16 floors ~~-. The simulation contributed only~~
1523 ~~to the new family; learned models matched the nearest produced setting in accuracy~~
1524 ~~and added calibrated abstention where the calibration alternatives resembled the new~~
1525 ~~design~~in two transfers to a new family. Within a family the choice of rule mattered
1526 more than the relation, release-level requirements lay inside every decisive distance,
1527 and in this small-data regime the simulation as used, a model with a large and partly
1528 sign-inconsistent bias, contributed only across families, while learned models added
1529 abstention rather than accuracy.

1530 Four extensions follow: series produced on several days, coils and tool states, so
1531 that the floor covers variation between runs; design families with continuous geomet-
1532 ric descriptors, ~~so that new designs can be placed between produced ones; a cost~~
1533 ~~model that turns the distances into decisions on trials and margins; a confirmatory~~
1534 ~~evaluation of the frozen procedure on another dataset or process;~~ and an evaluation of
1535 ~~the procedure~~ with design teams of whether decisions taken with the framework are
1536 better. The released code, which reproduces every result from the public data, is the
1537 starting point for all four.

1538 **Supplementary information.** The Electronic Supplementary Material contains
1539 Tables S1–S30 and Fig. S1 with the further results referred to in the text.
1540

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1542 datasets at the University of Stuttgart for publishing the data on which this work rests.
1543

1544 Declarations

1545
1546 **Funding.** No specific funding was received for this work.

1547 **Competing interests.** The author declares no competing interests.
1548

1549 **Ethics approval and consent.** Not applicable.

1550 **Data availability.** All data are public. RDDAC: doi:10.18419/DARUS-5589 (ver-
1551 sion 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0.
1552 The extracted per-part and per-simulation feature tables are provided with the code.
1553

1554 **Code availability.** The complete analysis pipeline is released under an open licence
1555 at <https://github.com/htcanseven/Claude> (directory `scripts/`; ~~an archived release~~
1556 ~~with a DOI will be deposited on Zenodo at acceptance~~) and reproduces every number
1557 in this paper from the public data; a versioned archive of the code and the feature
1558 tables is deposited on Zenodo [DOI pending].

1559 **Author contributions.** H.T.C. conceived the method, designed and carried out
1560 the analysis, and wrote the paper.
1561

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Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction as an assistant for code development, literature retrieval, data analysis and drafting and editing of the text, and for simulated peer review of earlier versions. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A ~~Computation details~~ Notation and additional tables ~~reuse~~

A.1 ~~Computation details~~

~~*Feature extraction.* Scans are reduced to a grid of 2×4 pixel medians after a luminescence filter (connected components of at least 20,040 pixels); calibration 0.0769 mm per pixel across the scan line, 0.1581 mm per line and 0.00775 mm per height unit. The cup-bottom plane is fitted robustly to the plateau at least 10 mm inside its edge, and all depths are taken from it. The flange comprises the points 20–40 mm below the bottom plane, at least 8 mm from the cup wall and 3 mm from the outer edge. Wall positions are interpolated where the binned (0.5 mm) radial profile reaches 10, 15 and 20 mm depth; the wall angle is taken between 10 and 20 mm. The arm angle is fitted to the longest run of the profile whose slope stays below 15° , shortened by 1 mm at both ends and at least 4 mm long. Flange widths are medians over bands ± 2 mm about the mid-lines; corner tips are the outline points within 0.3 mm of the extreme diagonal projection. Simulated parts are mirrored from the quarter model and processed with the same definitions on a 1 mm grid.~~

~~*Floors and effects.* Batches of $B = 50$ consecutive parts; batches with fewer than 80% valid values are dropped; centres are 10% trimmed means; $\alpha = 0.95$. Intervals: 2000 resamples of the batches within alternatives (seed 1), pairs of copies of one batch excluded, and 2000 resamples of whole alternatives. Scatter floor: 50 random permutations of the parts within each series. Components: rational subgroups of five consecutive parts; successive difference standard deviation $\sqrt{\Delta^2/2}$.~~

~~*Decision rules.* Adequacy share $p = 0.95$; per-geometry floors; requirements below zero are left out for these non-negative characteristics. The largest residual margin is the $\lceil (n+1) \cdot 0.9 \rceil$ -th smallest absolute residual of n calibration alternatives, the largest when this exceeds n . M5 and M5n: Gaussian-process regression with a constant-times anisotropic squared-exponential kernel plus white noise, normalised targets, three optimiser restarts (random state 0), length scales in $[0.1, 100]$; the noise level is bounded below by the mean squared block-bootstrap standard error of the calibration alternatives' q_{95} relative to the variance of the targets; descriptors are a convex-geometry indicator, the blank-holder force in units of 100 kN and the lubrication rank (coarse 0, medium 1, fine 2, the order of the oil amount), with descriptors that are constant over the calibration set dropped. NN: mean q_{95} of the calibration alternatives of the target's geometry at the nearest blank-holder force (both neighbours when equally near; the other geometry when the family has none). Me: for~~

each lubrication pattern, the friction coefficient of the DDACS grid at 0.99 mm whose simulated mid-side and corner draw-in, in floors, are closest in least squares to the centres of the calibration alternatives with that pattern (all calibration alternatives if none has it). M3 and M4: histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) on the geometry indicator, blank-holder force, two lubrication indicators, sheet thickness and oil film; 500 incoming conditions are drawn for the new alternative from calibration parts with the same lubrication pattern, with in-sample residuals pooled over the calibration alternatives resampled; offset and margin from a leave-one-out over the calibration alternatives.

Distances and intervals. The shares of correct and wrong verdicts are evaluated exactly on a grid of 0.05 floors up to 600 floors. Intervals: 1000 resamples of the held-out alternatives within each geometry (seeds derived from the scope), each with one of 1000 block-bootstrap replicates of every alternative's q_{95} (batches of 50 resampled within the series); the resamples are shared by all rules of a scope. Refitting bootstrap: 200 resamples of the alternatives within each geometry, every alternative-level rule refitted.

Budget, guard, scenarios, robustness. The budget uses all subsets of $k = 1, \dots, 8$ calibration alternatives of the family (M5 [Table A1 summarises the notation and the terms defined in Section 3](#), and M5n from $k = 2$), bands from 200 resamples of the held-out alternatives. The guards and the tolerance scenarios use the stored intervals of every rule. ISO 2768-1 angular rows by the shorter leg: the wall (about 28 mm) and the concave arm (about 18 mm) over 10 up to 50 mm, the convex arm (about 9 mm) up to 10 mm. The temperature correction subtracts $b(T - \bar{T}_a)$ from each characteristic, with b the pooled within-alternative slope on the punch temperature T . The smoothed simulation replaces the simulated values of each geometry and force by a quadratic response surface in sheet thickness and friction coefficient.

Software. Python 3.11 with NumPy 2.4, pandas 3.0, SciPy 1.17, scikit-learn 1.9 ([Pedregosa et al, 2011](#)) and Matplotlib 3.11.

A.1 Additional tables

Table ?? gives the bootstrap intervals of the decisive and safe distances, Table ?? the one-sided safe distances, Table ?? the coverage and width of the interval rules, Table ?? the paired differences between rules, Table ?? the distances per characteristic, Table ?? the part-level rules under other learners, Table ?? the hyperparameters of the Gaussian-process fits, Table ?? the redundancy of the scans and Table ?? the extraction noise of the simulations. Two analyses are provided with the code but not used here: the sensitivities of the simulation over the geometric corners of the simulation dataset, which have no production counterpart, and the in-line verifiability of the characteristics from the press-force record [Table A2 maps the framework's requirements onto other processes. Computation details are given in the ESM.](#)

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Table A1 Notation and terms

Symbol, term	Meaning
$a, c, q_p(a, c)$	alternative (design–process combination), quality characteristic, p -quantile of the parts of a ($p = 0.95$)
R, d	requirement (upper limit on c or $ c $); its distance from the truth in floors, $R = q_p + dF_c$
$[L_r, U_r], V_r$	interval of rule r for q_p ; verdict meets, fails or trial (Eq. 1)
B, m, F_c	batch size (50); batch centre (10 % trimmed mean); production floor (Eq. 2)
σ_b, σ_w	standard deviation between batch means and between parts within a batch
ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/ s_c $
u, v	exclusion distances of an interval from the truth, from above and below, in floors
$\kappa(\delta), \omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $ d = \delta$
δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
FA, FR side	one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects)
g	guard band: observable margin beyond which ≥ 95 % of decided verdicts were correct
Calibration alternatives	produced alternatives whose record calibrates a rule
Evidence relation	sibling (same process setting produced), new process setting (interpolation, extrapolation), new family
Calibration budget	distances against the number k of produced alternatives, by relation
Guard	family, range and novelty checks that send a design outside the calibration to trial

Table A2 What the framework requires, mapped onto other processes

Process	Batch and run	Dominant variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift within a run, coil and die state between runs	variant, process setting, family	batch centres within a run
Injection moulding	consecutive shots of one cavity; one mould set-up	cavity, shot, set-up	cavity, process window, new mould	nested: within cavity, between cavities
Machining	consecutive parts between offsets	tool wear, compensated by offsets	tool path, cutting data, new part	between offset intervals
Casting	castings of one pattern and melt replicated	melt, pattern position	alloy, gating, new pattern	between melts
Additive manufacturing	specimens of one build	build, position, orientation	orientation, parameters, new geometry	between builds, from replicated anchors

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